

REPORT : MAJOR PROJECT 2025-26	
PROJECT NAME(Group Number)	CloudCam Rover (Group Number 28)
DEPT OF ELECTRONICS AND TELECOMMUNICATION ENGINEERING	
 Atharva College of Engineering	
Students Names (Class /Roll Number)	SALONI SORAP (ET2/38)
Students Names (Class /Roll Number)	SANIKA UTEKAR (ET2/43)
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Students Names (Class /Roll Number)	VARAD PISE (ET2/46)
Mentor Name	PROF.PRAJAKTA PAWAR
SEM/Year/CAY	VIII/BE/2025-26
Domain	AI + IoT based Surveillance System
Project Name	CloudCam Rover
Problem Statement (Initial Goal)	The Cloud Cam Rover is an AI-powered surveillance vehicle capable of autonomous navigation. It integrates real-time video streaming, gas sensing, and detection. This project aims to create a mobile, intelligent, and eco-friendly security solution for both civilian and defense use.
OBJECTIVE(s)	1) Build an eco-friendly, self-driving surveillance rover 2) Enable video streaming with detection 3) Develop a smart solution for remote monitoring
SPECIFIC:	The goal of this project is to develop an autonomous surveillance rover capable of live streaming, detection. Our team includes 4 members (Saloni, Sanika, Om, Varad) who are responsible for the whole project. The project is divided in following tasks are assigned to each member 1) Identification, Procurement of components and design: Saloni, Sanika, Om, Varad 2) Simulation of circuit using any open source software : Om, Varad 3) Programming: Saloni, Sanika 4) Simulation : Varad 5) Fault finding and correction in both circuit & simulation: Saloni, Sanika, Om, Varad 6) Making of One page Report, synopsis, PPT: Saloni, Sanika
MEASURABLE:	The project will deliver a fully functional surveillance rover equipped with live video streaming, detection, and sensing. It will have the capability to store footage on the cloud and operate effectively. The rover will run on batteries, ensuring portability and eco-friendliness.
ACHIEVABLE:	All required components such as ESP32-CAM, gas sensors, and batteries are easily available and cost-effective. The team has experience with Python, OpenCV, and basic hardware integration, making implementation feasible. Tasks are divided among team members to ensure timely and effective completion of the project.

RELEVANT:

The Cloud Cam Rover addresses the need for intelligent, mobile surveillance by integrating AI, IoT, and cloud technologies for real-time monitoring. It offers a scalable solution for both civilian and defense applications while emphasizing cost-effectiveness and eco-friendliness. This project is highly pertinent given the increasing demand for advanced, adaptable security systems in a rapidly evolving tech landscape.

TIME-BOUND:

- **Week 1–2 (July 2025):** Orientation, understanding surveillance system concepts, and finalizing project topic “Cloud Cam Rover” with initial planning.
- **Week 3–4 (July–August 2025):** Conducted hardware research, prepared component list, and completed literature review on similar surveillance systems.
- **Week 5–6 (August 2025):** Procured components and assembled the basic surveillance car; tested motor control and camera module.
- **Week 7–8 (August 2025):** Performed motion testing and implemented basic live video streaming functionality.
- **Week 9–10 (September 2025):** Integrated movement with live video feed and debugged system for smooth operation.
- **Week 11–12 (September 2025):** Completed full system testing and verified real-time monitoring capability.
- **Week 13–14 (October 2025):** Prepared initial documentation and performed final testing of base system.
- **Week 15–18 (October–November 2025):** Added and tested gas sensor module for environmental monitoring.
- **Week 19–22 (November–December 2025):** Implemented detection system using camera feed for intelligent surveillance.
- **Week 23–26 (December 2025–January 2026):** Integrated detection, gas sensor, and rover movement; tested combined functionality.
- **Week 27–30 (January–February 2026):** Conducted multiple trials and simulated surveillance scenarios; improved system performance.
- **Week 31–34 (February–March 2026):** Final system integration, testing, and optimization of live streaming and detection.
- **Week 35–38 (March–1 April 2026):** Completed documentation, presentation, and final demonstration of the simulated surveillance car with gas detection and real-time monitoring system.

Introduction: The *Cloud Cam Rover* is a smart surveillance vehicle designed for real-time monitoring and data collection. It integrates live video streaming with intelligent detection capabilities for enhanced security. The system is equipped with a gas sensor to detect harmful environmental conditions. Built using components like the ESP32-CAM and motor control modules, it ensures efficient mobility and monitoring. The rover supports remote surveillance through wireless connectivity and real-time video feed. It has been tested under simulated conditions to demonstrate its performance and reliability. This project provides a portable, cost-effective, and efficient solution for modern surveillance applications

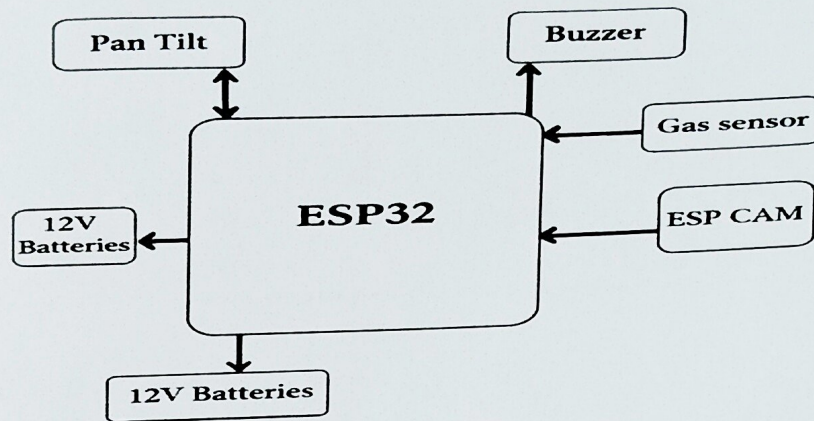
Description:

- The ESP32 CAM module also processes data from the motors and servos for coordinated control.
- The buck converter steps down voltage to power the servo motors efficiently.
- The pan and tilt servos adjust the camera to capture a wider area for surveillance.
- Motor driver L298N receives signals from the ESP32 CAM to control motor direction and speed.
- The web interface allows real-time monitoring and control of the system remotely.

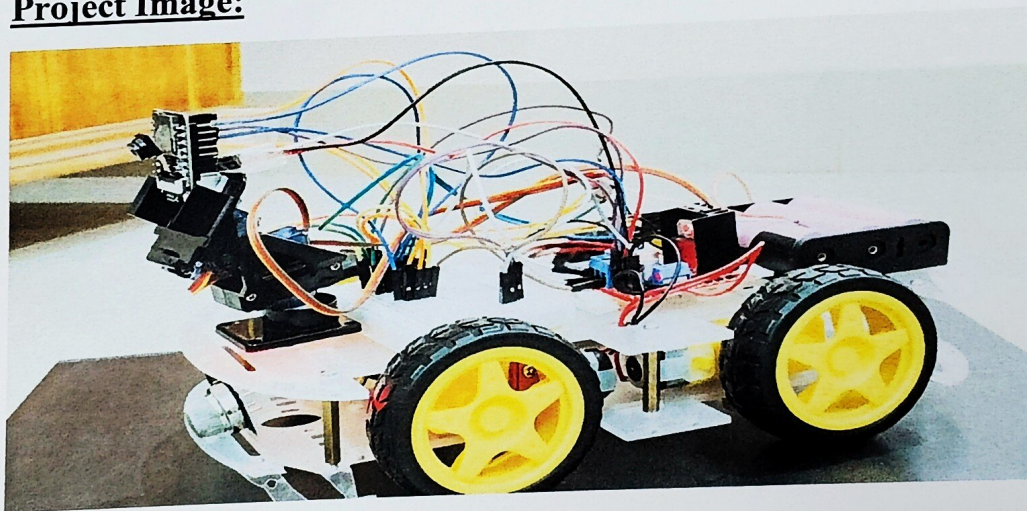
- The HTTPS protocol ensures secure transmission of the video feed to the user's web page.

All components are powered through a central power supply ensuring seamless operation.

Block Diagram:



Project Image:



Guide Name & Signature with date